Classes

1. Entrant (it's a class but it uses an interface called Entr) maybe we treat the profile as data that comes with every entrant instead of as a class on its own Done
2. CampsiteOwner (it's a class but it uses an interface called Owner) Done
3. Admin (this class uses Entrant and Owner interface) Done
4. Image(super class) Done
5. Campsite Done
6. QRCode Done
7. Location Done
8. Reservation: Mainly for Entrants to cancel reservation or choosing reservations Done
9. LotterySystem: to choose the five that get a lottery Done
10. RatingSystem : calculates average ratings Done
11. Ratings: the object ratings Done
12. OwnerView Done
13. WaitList Done
14. Message Done
15. AdminView Done
16. Notification Done
17. Notification Log Done
18. Profile Done
19. MessageLog Done
20. ProfileRepository Done
21. CampsiteRepository Done
22. CampImage (subclass) Done
23. ProfileImage (subclass) Done
24. CampView: each camp has a view showing its details including an option to join the lotteryDone
25. Screen Done
26. EntrantView Done
27. CalendarView Done
28. MapView Done
29. NotificationService Done
30. AvalabilitySchedule Done

Entrant	
1. Searches for campsite 2. Views/browse campsites and their details 3. Join active lottery for campsite 4. Join waitlist for campsite lottery 5. Leave waitlist for campsite lottery 6. Receives notifications from	1. Campsite 2. QRCode 3. LotterySystem 4. Rating 5. EntrantView 6. Screen 7. Reservation

Entrant	
LotterySystem/CampsiteOwner 7. Make or cancel reservations for campsites 8. Scan QR codes 9. Give ratings for campsites 10. Upload profile image 11. Browse campsites	8. ProfileImage 9. Notification 10. Message 11. CampView

CampsiteOwner	
1. Register their campsite(s) 2. Provide and update campsite details 3. Create location 4. View only their own campsite details through OwnerView 5. Generate QR code for their campsite 6. Set registration and booking period for lottery 7. Set waitlist capacity and modify lottery system for their campsite(s) 8. View past, active, and cancelled reservations for campsite 9. Send and view notifications to entrants that join waitlist 10. View entrants on waitlist 11. Message entrant reserving their campsite	1. Entrant 2. LotterySystem 3. QRCode 4. Location 5. Campsite 6. Reservation 7. Notification 8. Message 9. Screen 10. OwnerView 11. CampImage 12. CampView

Admin	
1. All responsibilities of Entrant 2. All responsibilities of CampsiteOwner 3. Remove campsites 4. Remove campsite owners 5. View all notifications through logs 6. Remove all images 7. Review all messages	1. Entrant 2. LotterySystem 3. QRCode 4. Location 5. Campsite 6. Reservation 7. Notification

Admin	
8. Browse profiles of all entrants 9. Browse all campsites	8. Message 9. Screen 10. AdminView 11. Image

Image	
1. Is uploaded by all actors 2. Can be deleted by admin 3. Displayed in views	1. Campsite 2. Screen 3. AdminView 4. OwnerView 5. EntrantView 6. CampView 7. Admin 8. Entrant 9. CampsiteOwner

LotterySystem	
1. Randomly picks n number of lottery winners for a campsite (n is determined by CampsiteOwner). 2. Sends list of winners of campsite to the campsite owner 3. Sends notification to winners that they won	1. Entrant 2. CampsiteOwner 3. Campsite 4. WaitList 5. NotificationService 6. NotificationLog 7. AvailabilitySchedule

Reservation	
1. Made and cancelled by Entrants 2. Can be viewed by all actors (to varying degrees) 3. Can be rejected by owners	1. Entrant 2. CampsiteOwner 3. Admin 4. AdminView 5. OwnerView 6. EntrantView 7. AvailabilitySchedule 8. Campsite

QRCode	
1. Scanned by Entrants to view campsite details 2. Created by owners for campsites 3. Shown on camp view	1. Campsite 2. Entrant 3. CampsiteOwner 4. EntrantView 5. OwnerView 6. CampView

RatingSystem	
1. Handles rating process 2. Calculates aggregate/average rating of campsite 3. Collects Ratings given by entrants	1. Campsite 2. Rating

WaitList	
1. Holds the entrants who joined the lottery per campsite 2. Adds an entrant when they join (prevents duplicates, and rejects those attempting to join after the wait period) 3. Removes an entrant when they leave 4. Holds the optional capacity set by campsite owners 5. Provides a list of entrants to the campsite owner 6. Provides a list of entrants to the lottery system for the draw	1. CampsiteOwner 2. Campsite 3. Entrant 4. LotterySystem 5. Notification 6. NotificationService

Rating	
1. Knows the rating value	1. Entrant

Rating	
2. Knows which entrant submitted which rating 3. Knows which campsite the rating applies to 4. Can be updated by entrant 5. Stored by the rating system	2. Campsite 3. Rating system 4. Reservation

Notification	
1. Knows the recipient (entrant) 2. Knows the sender (by the system or campsite owner) 3. Knows the type of notification (win, lose, reservation, cancellation, waitinglist) 4. Knows the related campsite, lottery or reservation 5. Sent only if entrant has not opted out 6. Knows the information and timestamp 7. Be logged so that campsite owners can see sent history and admin can review it	1. Entrant 2. CampsiteOwner 3. Campsite 4. LotterySystem 5. Reservation 6. NotificationLog 7. Admin 8. NotificationService

Message	
1. Knows the sender (campsite owner) 2. Knows the recipient(entrant) 3. Knows which reservation it relates to 4. Knows message time stamp 5. Keeps track of it read/unread status 6. Logged so that owners can see history and admins can review it	1. Entrant 2. CampsiteOwner 3. Reservation 4. MessageLog 5. Admin 6. NotificationService

Campsite

1. Knows the name, description and price 2. Knows the maximum camp occupancy 3. Knows the rules, regulations and policy 4. Knows it owner 5. Knows it's location 6. Knows the QR Code link to it 7. Knows the rating related to it 8. Knows its availability Schedule 9. Knows which reservation it relates to 10. It can be created, updated and removed by campsite owner or admin 11. Provides summary information 12. Provides full details	1. CampsiteOwner 2. Location 3. QRcode 4. CampImage 5. RatingSystem 6. Reservation 7. CampView 8. CalendarView 9. CampsiteRepository 10. AvailabilitySchedule

Location

1. Knows latitude, longitude and a proper address 2. Set by owner when registering camp 3. Provides coordinates for map 4. Supports location filtering (by distance)	1. Campsite 2. CampsiteOwner 3. MapView

Profile

1. Knows name, email and phone number of user 2. Knows the notification preference 3. Knows the uploaded profile image 4. Updated by entrant 5. Can be deleted by entrant or admin 6. Can be explored by Admin	1. Entrant 2. EntrantView 3. ProfileImage 4. Admin 5. ProfileRepository

NotificationLog

1. Captures the details of all notifications sent (sender, recipient, content, timestamp, etc)
2. Receives log entries from the LotterySystem (the winners/ losers of the draw)
3. Provides the full list of notifications to the admin for review
4. Provides the CampsiteOwner with the notifications they sent
5. Keeps entries even after an entrant opts out (potentially unnecessary)

1. Notification
2. LotterySystem
3. CampsiteOwner
4. Admin
5. Entrant
6. Campsite

MessageLog

1. Records all the messages a CampsiteOwner sends to an entrant (sender, recipient, content, timestamp, campsite)
2. Receives log entries from NotificationService when a message is sent
3. Provides a full list of messages to the admin for review
4. Provides the messages for a given reservation to the CampsiteOwner and entrant involved
5. Keeps a record of the entries after a reservation is ended or cancelled

1. CampsiteOwner
2. Entrant
3. Campsite
4. Message
5. Admin

OwnerView

1. Allows rejecting a reservation
2. Shows QR code for each campsite
3. Shows waitlist entrants for a lottery
4. Shows the register campsite forms

1. CampsiteOwner
2. Campsite
3. LotterySystem
4. WaitList
5. Reservation
6. QRCode
7. Notification Log

OwnerView	
5. Shows just the owners campsite and the details 6. Shows lottery info with registration period,booking period, waitlist capacity and winner amount 7. Shows the active, cancelled and past reservations 8. Shows messaging screen with sent notifications	8. Message 9. MessageLog 10. CampImage 11. CalendarView 12. MapView 13. CampView

AdminView	
1. Lets admin switch between entrant and owner view in their own profile 2. Shows notification and message logs 3. Lets admin remove all campsite, profiles and images 4. Provides action to remove campsites, its owners, images and profiles	1. Admin 2. Campsite 3. Profile 4. Image 5. NotificationLog 6. MessageLog 7. Campsite Repository 8. ProfileRepository

ProfileRepository	
1. Provides all profiles for admin to browse 2. Looks up a profile by device ID 3. Removes profile and profile image when deleted 4. Handles saving, updating, deleting and retrieved profiles from Firebase	1. Profile 2. Entrant 3. Admin 4. Profile Image

CampsiteRepository	
1. Returns an owners campsite 2. Provides all campsites for admin to browse 3. Handles search and filter queries 4. Handles saving, updating, deleting and retrieved campsites from Firebase	1. Campsite 2. CampsiteOwner 3. Admin 4. Entrant 5. AvailabilitySchedule

CampImage	
1. Displayed on campsite listings and details 2. Can be removed by owner or admin 3. Knows which campsite it belongs to 4. Uploaded by campsite owner as promotional image	1. Image 2. Campsite 3. CampsiteOwner 4. Admin 5. OwnerView 6. EntrantView

ProfileImage	
1. Displayed on the profile screen 2. Knows which profile it belongs to 3. Can be removed by entrant or admin 4. Its uploaded by entrant	1. AdminView 2. Admin 3. Entrant 4. Image 5. Profile 6. EntrantView

Screen	
1. Defines show, refresh and navigate 2. Holds currents users role 3. Shows images	1. EntrantView 2. OwnerView 3. AdminView 4. Image 5. CampView 6. CalendarView 7. MapView

CampView	
1. Shows campsites full details 2. Opens from search results or from QR code 3. Shows the campsite QR code 4. shows the option to join or leave the waitlist when a lottery is active for entrants 5. shows the option to edit the campsite and manage the lottery for the owners 6. shows the option to remove the campsite for admins	1. Campsite 2. Entrant, CampsiteOwner 3. Admin 4. WaitList 5. LotterySystem 6. QRCode 7. CampImage 8. CalendarView 9. MapView 10. RatingSystem

EntrantView	
1. Shows the search and filter screen (location, date, number of guests) 2. Shows the list of campsites based on the filters, with summary info (name, location, maximum occupancy, rating) 3. Launches the QR scanner and opens the CampView 4. Shows the lottery result screen (choose campsite and dates if the entrant won, or decline) 5. Shows the entrant's active and past reservations, with the option to cancel	1. Entrant 2. Campsite 3. Reservation 4. WaitList 5. LotterySystem 6. QRCode 7. Location 8. RatingSystem 9. Profile 10. ProfileImage 11. CampImage 12. Notification

EntrantView	
6. Shows the rating screen for a campsite 7. Shows the profile screen (edit info, upload profile image, delete profile, notification opt-out) 8. Shows the app policy and guidelines 9. Shows received notifications and messages	13. Message 14. AvailabilitySchedule 15. Screen 16. CampView 17. CalendarView 18. MapView

CalendarView	
1. Displays campsite availability with date 2. Lets owners set availability dates 3. Lets entrants pick from available date when reserving 4. Marks the booked dates	1. Campsite 2. AvailabilitySchedule 3. Reservation 4. CampView 5. EntrantView 6. OwnerView

AvailabilitySchedule	
1. Knows which dates the owner made available 2. Knows which dates are already booked 3. Blocks a date when a reservation is confirmed and unblocks it when reservation is cancelled or rejected 4. Answers if campsite is free on a certain date for filtering and booking	1. Campsite 2. Reservation 3. CalendarView 4. CampsiteRepository 5. LotterySystem

MapView	
1. Displays a map on a campsite's coordinates 2. Reads latitude and longitude from Location 3. Lets the owner pick or confirm the location when registering or editing	1. Location 2. Campsite 3. CampsiteOwner 4. Entrant 5. CampView 6. OwnerView

NotificationService	
1. Delivers notifications and messages to entrants 2. Checks the entrant's preferences for opting out before sending 3. Sends win/lose notifications from LotterySystem and custom notifications from owners 4. Sends reservation and cancellation confirmations 5. Helps with writing send to NotificationLog and MessageLog	1. Notification 2. Message 3. NotificationLog 4. MessageLog 5. Profile 6. Entrant 7. CampsiteOwner 8. LotterySystem